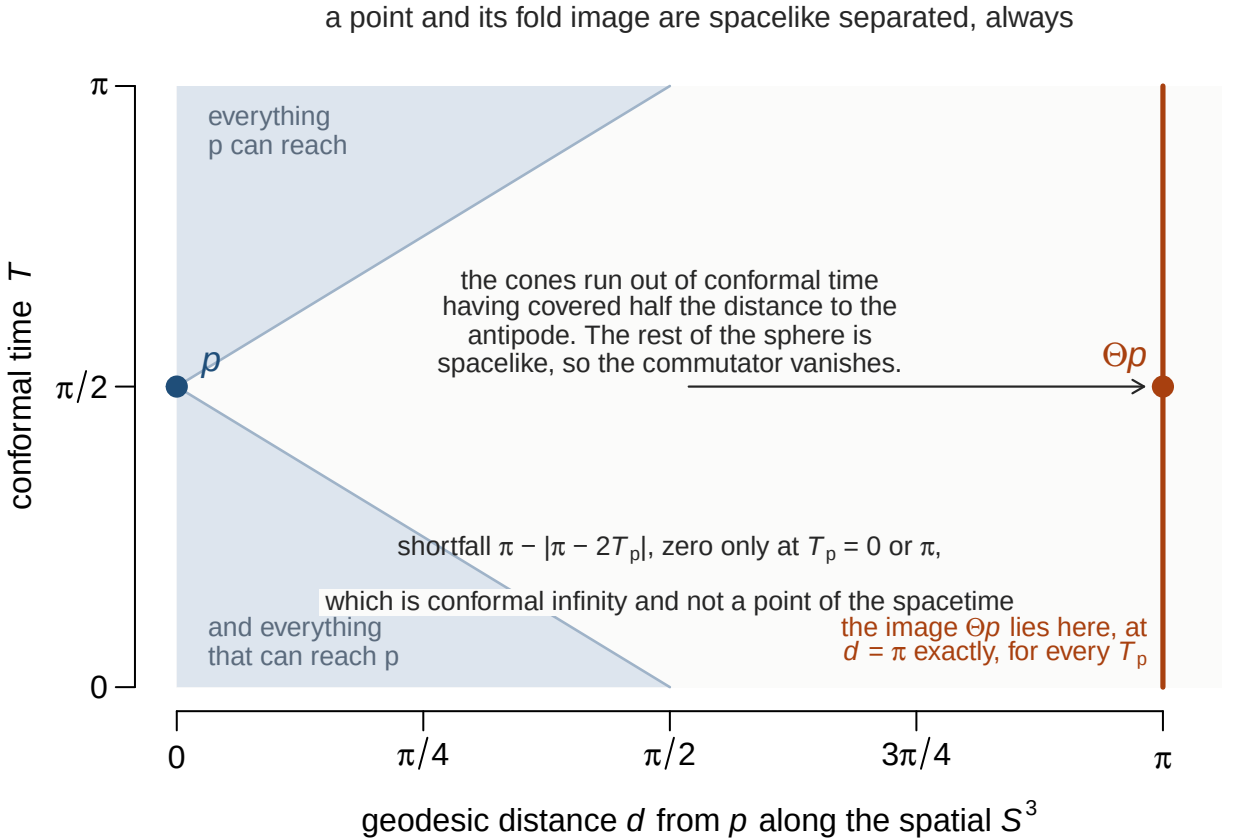
a point and its fold image are spacelike separated, always
π
π/2
0
conformal time T
everything
p can reach
p
the cones run out of conformal time
having covered half the distance to the
antipode. The rest of the sphere is
spacelike, so the commutator vanishes.
Θp
shortfall π − |π − 2T_p|, zero only at T_p = 0 or π,
which is conformal infinity and not a point of the spacetime
and everything
that can reach p
the image Θp lies here, at
d = π exactly, for every T_p
0
π/4
π/2
3π/4
π
geodesic distance d from p along the spatial S^3